Proving Ground

Make anything.
Watch a real robot deal with it.

Aholo Lux3D turns a photo or a sentence into a physical object. Unitree's published G1 walking controller has to get past it, live in the browser.



proving-ground-snowy.vercel.app

Manycore Aholo Global 3D AI Hackathon · Rohan Borade (solo) · November 2026

Robots fail on the things nobody modelled

Walking and grasping controllers are tested on a handful of hand-made shapes. Real floors have shoes, toys, cables, chairs and bags.

Lux3D can build almost any object, at its real size, in about 80 seconds, from a sentence or a photo.

So: a robot test range where every obstacle is made by Lux3D, and the robot is a real controller, not an animation.

Robotics teams

Varied, realistic test objects with real sizes and collision parts in 90 seconds, downloadable as MuJoCo-ready packs.

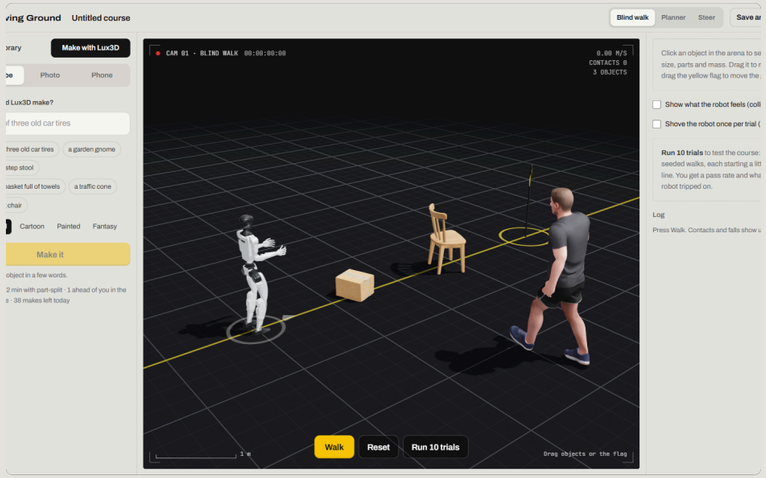
Anyone curious about robots

Make something from your own life and watch a real controller deal with it. Share the course that beat it.

Lux3D as an asset factory

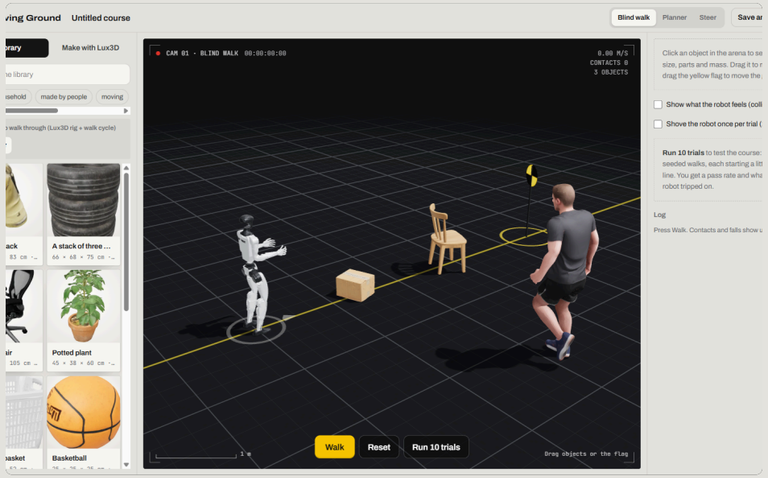
Shows Lux3D output going straight into a physics simulator: size, parts, people and motion, no hand modelling.

What you can do in one minute



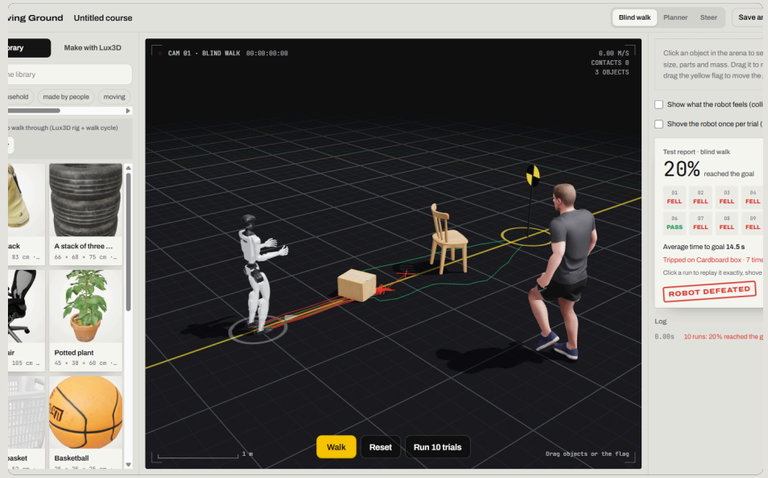
Make it

Type a sentence, upload a photo, or scan a QR code and snap with your phone.



Drop it in

It materializes at its real size in front of the robot. Drag it, turn it, change its density.



Test it

Walk once, steer, or run ten seeded trials for a pass rate and what it tripped on. Share the course.

Every object passes through Lux3D

Step	Aholo call	Credits	Time
Upload a photo or GLB	Asset upload (OUS)	0	1–3 s
Photo to 3D	img-to-3d · G1-Turbo	1	~80 s
Sentence to 3D	text-to-3d · G1-Turbo	13	~80 s
Split into parts	part-split	30	~80 s
Rig a person	animations/rig	15	~63 s
Make them walk	animations/retarget · Walk_Loop	6	~120 s
Price every job first	pricing/openapi-quotes	0	<1 s

The library right now, all made by Lux3D

18 objects and people

52 collision parts

7 split by part-split

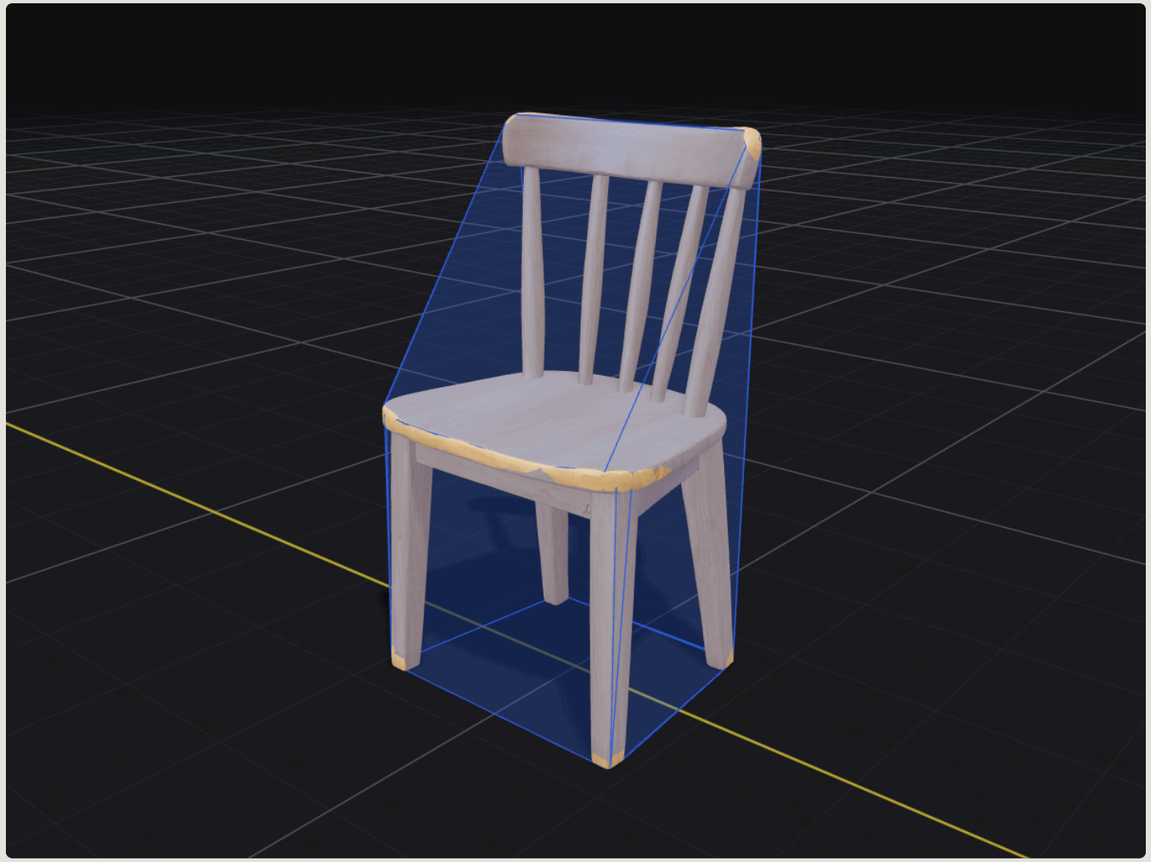
1 rigged walkers

Inputs must go through Aholo's own upload: third-party image URLs fail silently, so every photo and every intermediate GLB is re-uploaded.

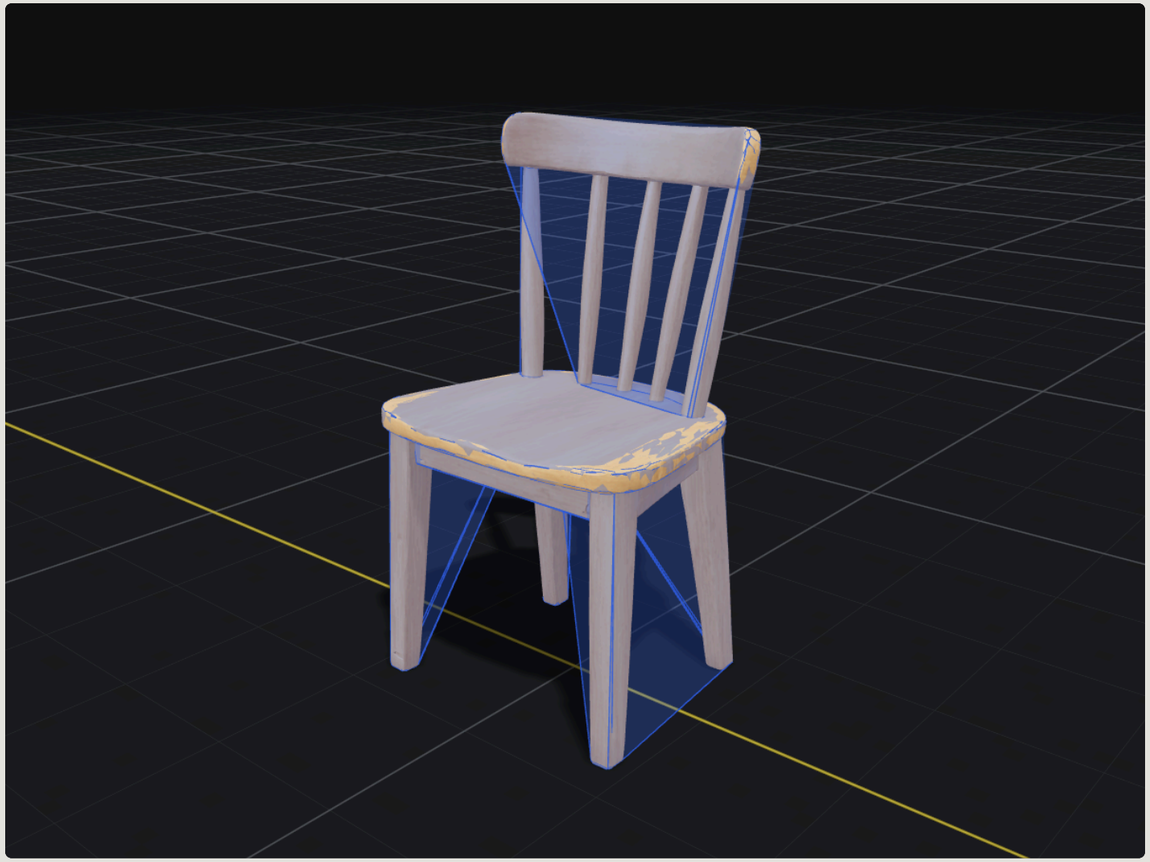
Lux3D runs one task per account, so a single queue serves everyone; the library keeps the site playable while it is busy.

Lux3D predicts real size: a sneaker comes back 28 cm long and a chair 85 cm tall with no one typing a dimension.

Iteration 1: one block, or the parts it is made of

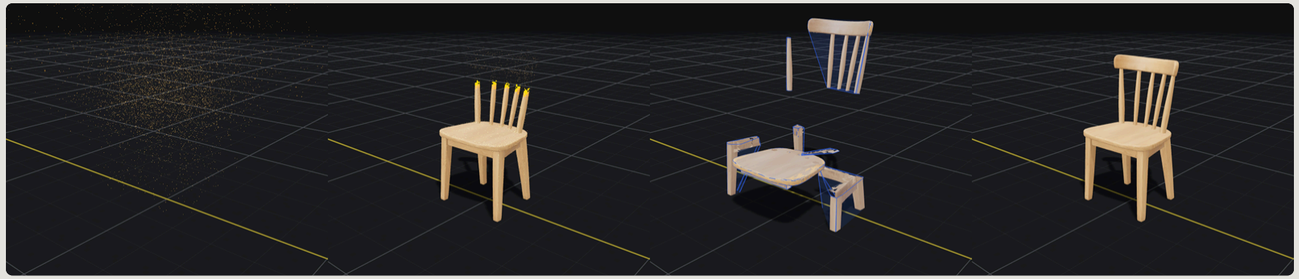


BEFORE Lux3D mesh as one hull: the chair is a solid block. Feet cannot go between the legs.



AFTER Lux3D part-split: eight solids (legs, seat, back). Contacts name the part the robot hit.

Iteration 2: real size, honest mass, light files



Choice	First try	Now
Collision	1 hull per object	1 hull per Lux3D part
Chair mass	18.8 kg (hull volume)	5.2 kg (closed mesh volume)
Face count	300k faces, 19 MB	20k faces, ~7 MB from Lux3D
Web file	raw GLB	simplified + compressed visual
Inputs	public image URL	Aholo asset upload

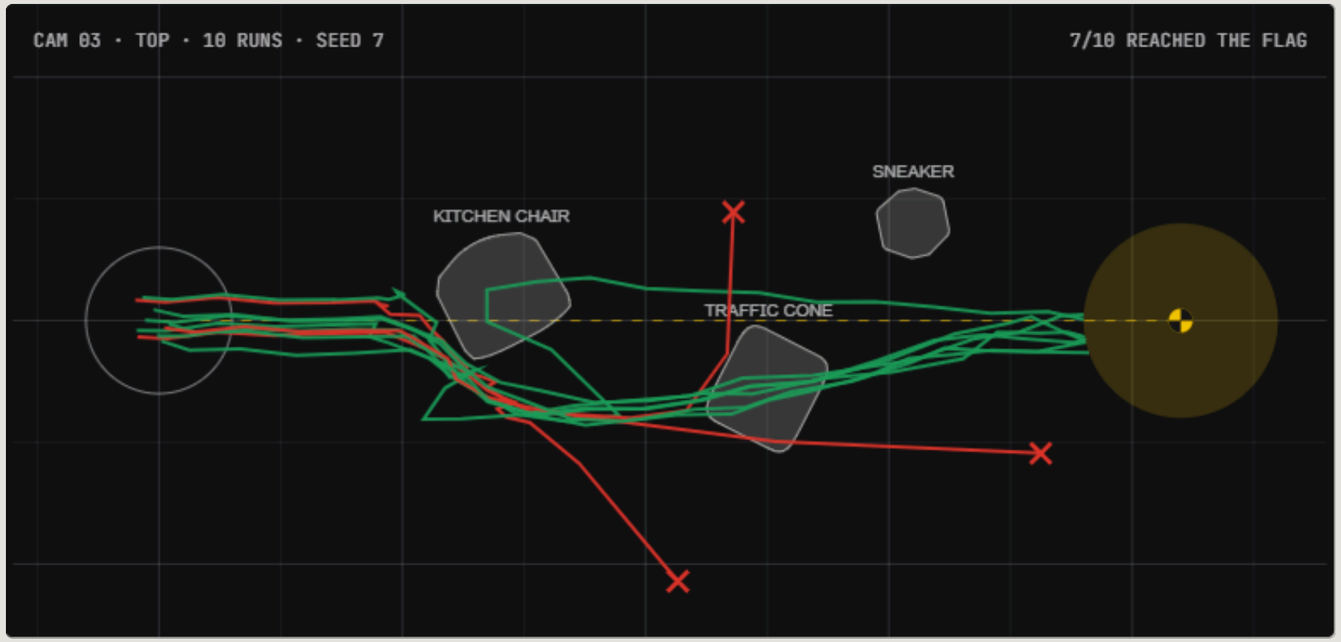
The robot is real software

Unitree's published G1 walking policy (unitree_rl_gym, BSD-3): an LSTM read straight from the TorchScript file and run in plain JavaScript at 50 Hz on 500 Hz MuJoCo physics.

It is blind: it only feels its joints. Planner mode adds a map of object footprints and an A* path, so the difference between the two pass rates is the value of perception.

Gap to Unitree's Python run after 2 s	**2.5 µm**
Walk 6 s at 0.5 m/s	**2.6983 m (same as Python)**
Speed in the browser	**12× real time (one chair), 3–4× (20 objects)**
10 trials of a chair course	**3.7 s wall time**
Compile a 20-object scene	**163 ms**
One 250 N shove at a random moment and angle	**falls 5 of 10 walks**

Ten runs, not one



A Lux3D park bench across the path: blind walk reaches the flag 0 of 10 times (it pushes a 56 kg bench and stalls); the planner routes around it 10 of 10. That gap is what perception is worth, measured.

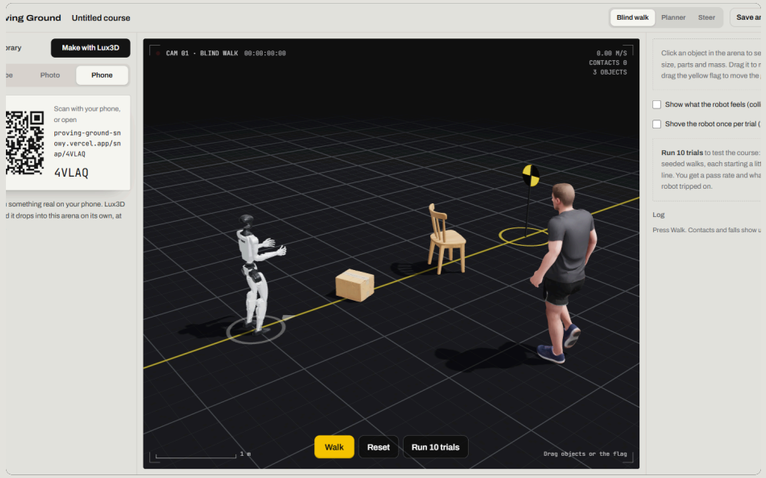
Trip test of the 8-part Lux3D chair, 10 blind walks: 5 cleared, 1 fall, 4 stuck pushing it. The 0.55 kg sneaker and the 2.7 kg cone are cleared 10 of 10.

A kitchen-floor course (chair, sneaker, cone): 8 of 10 blind walks reach the flag.

The plot: chair, cone and sneaker with one 170 N shove at a random moment in each run. 7 of 10 reach the flag; the red crosses are where the other 3 fell.

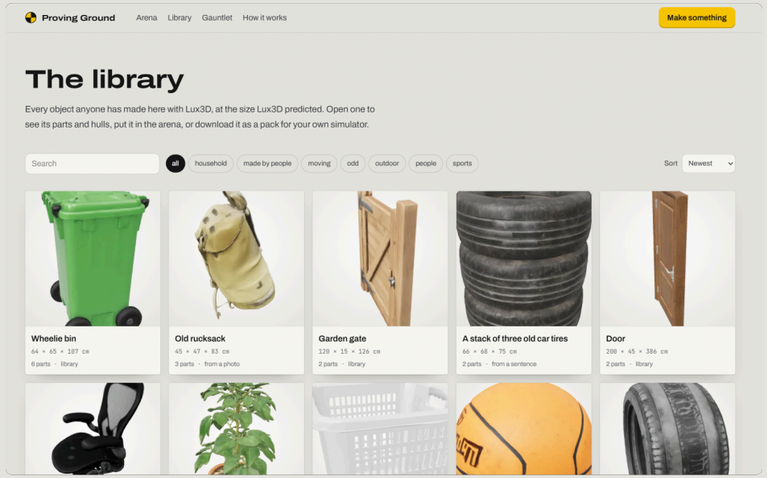
Fixed 6 cm and 12 cm steps trip the robot every time, and every trial records what it tripped on.

Built for people to play with



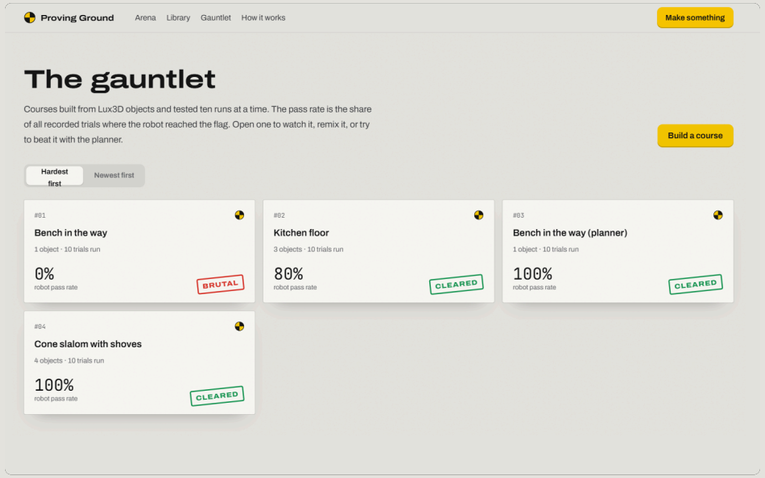
Snap from your phone

Scan the QR code, photograph something real, and it lands in the desktop arena.



A shared library

Every object anyone makes, with parts and hulls, downloadable as a robot-ready pack.



The gauntlet

Courses ranked by how often the robot fails them. Remix one, or beat it with the planner.

What it cannot do yet, and what is next

- The controller is blind and was trained on flat ground; it never sees the objects.
- Each part collides as its convex hull; hollow parts (a bowl) are filled.
- Mass is an estimate from volume and a density you can change.
- Walking people are kinematic: they push the robot but cannot be pushed.
- One Lux3D generation at a time per account.

- Grasping: the same objects for a manipulation policy (Lux3D part-split gives handles and lids for free).
- Lux3D world generation as the floor plan: test inside a generated room, not on an empty floor.
- Export courses as MuJoCo and Isaac scenes for CI on real robot code.

Try it: **proving-ground-snowy.vercel.app**
Source: github.com/RohanGlitched/proving-ground
Video: https://youtu.be/6pRu4K838hk